

User Stories

Coding Blocks

- Mechanics & Structure:** As a user, I want a 'base block' system where I can drag and drop commands into a sequence to build a script.
 As a user, I want block to visually connect and 'snap' to each other only when valid, so I can easily build chains of code without worrying about syntax errors.
 As a user, I want each block to be highlighted during execution, so I can trace exactly which block is executing at any moment.
 As a user, I want to be able to undo/redo my last action when creating my code.
- Action Blocks:** As a user, I want specific movement, jump and action blocks so I can navigate the player character through the level.
 As a user, I want a 'wait' block with a customisable duration or condition so that I can pause the character's actions for a specific time.
- Logic & Conditionals:** As a user, I want 'if statement' blocks that can wrap around other blocks to make the character perform actions only when certain conditions are met.
 As a user, I want specific 'sensor check' blocks so my code can react to the environment.
 As a user, I want 'while loop' and 'for loop' blocks that repeat a set of actions depending on conditions.
- Debugging & Safety:** As a user, I want to see a text log in a 'console' for each block executed, so I have an easy-to-access log for debugging.
 As a user, I want the console to show if an action failed or succeeded, so I can debug my code.
 As a user, I want the system to automatically detect and stop infinite loops so the application doesn't freeze if I make a logic error.
- Advanced Features:** As a user, I want to be able to create 'function' blocks that combine multiple actions into one, allowing me to simplify complex code.
 As a user, I want to be able to create variable blocks to store data (e.g. a counter to track how many times an event has occurred).
 As a user, I want to be able to use my variables as inputs for other blocks, so I can write dynamic code that changes based on game events.

Graphics

- Visual style:** As a user, I want a clean, modern pixel art appearance to the game. • As a user, I want a distinct visual style (e.g. Fantasy, sci-fi, etc.) that is consistent throughout the game and wider application.
- Colour palette:** As a user, I want a modern colour palette/'dark mode' so that the application feels mature, rather than childish.
- Code block appearance:** As a user, I want 'code blocks' to differ in shape and colour to distinguish function.
- Gameplay action indication:** As a user, I want to see which blocks can link with each other through some sort of visual indicator, like a 'puzzle piece' shape.

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- **Interface icons:** As a user, I want high-res vector icons for UI elements so that the interface can be resized/viewed on screens of different sizes.

Front-End

- **Dynamic interface:** As a user, I want the interface to automatically adjust to my screen resolution, so that it is never cut off or stretched.
- **Resolution scaling:** As a user, I want pixel art assets to scale appropriately for screens of different sizes/resolutions.
- **Orientation:** As a user, I want the application to force a landscape screen orientation so that I always have optimal view and can see all relevant elements.
- **Separation of UI concerns:** As a user, I want the 'code block' environment (UI) to remain separate from the game view.

Mechanics & Game Design

- **Player Control & State:** As a user, I want the player character to move, jump, turn only when my code commands (except for environmental factors, if relevant), so that I understand that the computer only does what I tell it to.
As a user, I need the character to have a specific 'facing direction' so that relative commands (e.g. turn left, move forward) make logical sense.
As a user, I want to navigate vertical space by jumping onto ledges, dropping down and moving under/over obstacles so the world doesn't feel flat.
- **Execution & Debugging:** As a user, I want a prominent 'execute'/'go' button to start my code so I can clearly separate the 'coding phase' from the 'running phase'.
As a user, I want the ability to pause the game mid-execution so I can analyse what is happening at a specific moment.
As a user, I want to be able to 'step through' my code one block at a time, so I can catch exactly where my code is going wrong.
As a user, I want a 'reset' button that instantly returns the character and level to their starting state, so I can quickly retry after a failed attempt.
- **Camera:** As a user, I want the camera to always follow the player character during execution so that I can see the result of my code, even in larger levels.

Level Design

- **Environment & Tiles:** As a user, I want clear visual distinction between 'safe' and 'hazard' tiles in levels so that I can plan my code to avoid them.
As a user, I want the game to include 'z-levels'/elevation, where some tiles are higher than others, requiring logic to determine when to jump, and how to navigate a '3D' environment.
- **Interactable Objects:** As the user, I want to encounter objects that block my path, but can be opened/moved/destroyed, forcing me to write additional code to overcome them.
As the user, I want collectible items that disappear upon contact, providing me with a score boost and optional challenge.
As a user, I want to interact with 'switches' that change the environment (e.g. opening a door, activating a moving object), requiring additional code to complete the level.

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- **Progression:** As a user, I want the first level to require only basic movement and sequencing to get me used to the interface and logic of the game without being overwhelmed.
As a user, I want specific levels designed to teach me 'conditionals', where I must detect a hazard and choose to bypass it.
As a user, I want levels that are tedious/difficult to solve with just 'move' blocks, motivating me to learn and use 'loop', 'function' and 'variable' blocks.

Sound & Audio Design

- **Basic Interaction Sounds:** As a user, I want a small "click" sound when I place a code block so I know the action registered.
As a user, I want a different "snap" sound when two code blocks connect properly so I know they are linked.
As a user, I want an error sound when blocks cannot connect so I immediately know something went wrong.
- **Player Movement Sounds:** As a user, I want a footstep sound when the character walks so I can tell they are moving.
As a user, I want a jump sound so I know the character performed the jump.
As a user, I want a landing sound so movement feels more realistic.
- **Gameplay Feedback Sounds:** As a user, I want a "success" sound when my code works correctly so I feel rewarded. As a user, I want a "fail/error" sound when my code breaks or the character can't complete an action. As a user, I want a completion sound when I finish a level so it feels like clear progress.
- **UI Sounds:** As a user, I want soft sounds for button clicks so menu interactions feel responsive. As a user, I want a hover/select sound when choosing options so I know where I am in the menu.
- **Environmental Sounds:** As a user, I want basic ambient sounds (wind, cave echo, sci-fi hum, depending on theme) to make the level feel alive. As a user, I want objects like doors, switches, or moving platforms to have simple sound cues so I understand when they activate.
- **Startup Sounds:** As a user I want sounds to play in the starting of the game, and when the main menu pops up, so I know things are working.

User Interface

- **Clean interface:** as a user I want an uncluttered interface so I can focus on the application without being distracted by unnecessary elements
- **Collapsible UI:** as a user I want to have control over how to show or hide parts of the interface to make better use of limited screen space
- **Consistent design:** as a user I want consistent colours and shapes for blocks and interactable elements
- **Visual hierarchy:** as a user I want to intuitively understand where the palette, workspace, and console are in order to look for tools, output, and results.

User Experience

- **Responsiveness:** as a user I want an application that loads quickly and does not lag when completing on any user action

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- **Feedback:** as a user I want immediate visual & audio feedback when I perform actions on the app
- **Locus of control:** as a user I want to feel like I'm in control and be able to undo/redo/paste blocks without worrying about mistakes
- **Accessibility:** as a user I want options that allow me to use the app regardless of my visual needs
- **Clear tutorialisation:** as a user I want tooltips and greying-out of unavailable features so I can understand what is available and to use it
- **Stepped execution:** as a user I want to be able to understand the step-by-step of executable actions
- **Clear error feedback:** as a user I want to immediately understand why something didn't work or I couldn't do something
- **Minimal memory load:** as a user I don't want to have to remember long sequences of actions or events, I want to have information presented logically.
- **Quick reset:** as a user I want to be able to clear or reverse my workspace to start over easily
- **Guidance:** as a user I want hints or suggestions on difficult challenges
- **Progression:** as a user I want visible achievements or metrics that help me feel like I'm being rewarded
- **Shortcuts:** as a user I want to be able to learn quick actions to navigate quicker as I get more proficient with the application

User Data and Social Connectivity

- **Progress sharing:** As a user, I want an online sync feature so I can share progress or compare scores with others if I choose.
- **Secure data:** As a user I want to trust that I'm not giving over any personal data that will be exposed